BCA303-5 - ADVANCED PYTHON

Course Plan

SECTION I

Semester	**5 BCA**	Class	**BCA (A) and (B) Sections**
Course Code	**BCA303-5**	Course title	**ADVANCED PYTHON**
Hours	**75**	Hours per week	**5(3+2)**
Faculty name	**Mr. Prateek Singhal**	Contact details	smera.c@christuniversity.in deepa.bg@christuniversity.in Mob: 8960061347 Email: prateek.singhal@christuniversity.in meenakshi.rajoriya@christuniversity.in smruti.dabhole@christuniversity.in
Credits	**4**	Course Type	**Major Core**
Class policies and guidelines	1. All communication will be done through the CHRIST University Official Mail/official class WhatsApp group. 2. Read the course content/resources shared in Google Drive before coming to the class. 3. Usage of the Laptop is only with the prior permission (as per the requirements of the class/Lab/demo) 4. Help / Discussion / Mentoring will happen after regular class timing		
Course Description	This course covers the advanced concepts associated with Python such as object-oriented programming, Graphical programming, and Web applications of Python with the help of built-in modules. This course aims to provide comprehensive knowledge of Python programming paradigms.		
Course Objectives	This course aims to provide comprehensive knowledge of Python programming paradigms.		
Learning Outcomes for the Course	CO1: Apply Object-Oriented Programming concepts and implement regular expressions for form validation. CO2: Design GUI-based applications using Tkinter or PyQt modules. CO3: Build interactive web applications using Streamlit and Flask frameworks. CO4: Analyze file handling operations and structured data using Pandas.		

SECTION II					
Unit number and title	**Unit details**	**Week (starting and end dates)**	**Hours per Chapter**	**Pedagogy (teaching learning methods used)/ activities and or class trips/ dates for assessment**	**Resource/ Reference details**
Unit-1 **OBJECT-ORIENTED PROGRAMMING USING PYTHON AND REGULAR EXPRESSIONS**	Classes: Classes and Instances-Inheritance	**Week 1** (07/07/25 - 12/07/25)	05	**Chalk and Board/ PPT Presentation** **Google classroom** **flippity**	Wesely J.Chun,Core Python Application Programming , Prentice Hall,third edition 2015. T.R.Padmanabhan, Programming with Python, Springer Publications, 2016
	Abstract class with concrete methods, constructors -Exceptional Handling	**Week 2** (14/07/25 - 19/07/25)	05	Chalk and Board/ PPT Presentation Google classroom	Wesley J.Chun,Core Python Application Programming , Prentice Hall,third edition 2015.
	Regular Expressions using “re” module, Common use cases: validation, parsing, searching	**Week 3** (21/07/25 - 26/07/25)	05	**Chalk and Board/ PPT Presentation** **Google classroom** **Class Discussion:**	Wesley J.Chun,Core Python Application Programming , Prentice Hall,third edition 2015.
Unit-2 **GUI PROGRAMMING**	Introduction-Tkinter/ PyQT module- Root window-Widgets	**Week 4** (28/07/25-02/08/25)	05	**Chalk and Board/ PPT Presentation Kahoot, Google classroom**	LMS Moodle/ Online Forum

	Common widget types: interactive and display -Button-Label-Styling	**Week 5** (04/08/25-09/08/25)	05	**CIA – I**	Wesely J.Chun,Core Python Application Programming, Prentice Hall,third edition 2015.
	Formatting labels-Message-Text-Menu-Listboxes-Spinbox- Table Layout.	**Week 6** (11/08/25-16/08/25)	05	**Chalk and Board/ PPT Presentation Kahoot, Google classroom**	Lab Test - I
Unit-3 **INTRODUCTION TO WEB FRAMEWORK**	Building and Deploying Streamlit Apps: Streamlit Introduction - Basic Streamlit setup, creating simple apps, and displaying text/output on a web page	**Week 7** (18/08/25-23/08/25)	05	**Chalk and Board/ PPT Presentation Kahoot, Google classroom**	Wesely J.Chun,Core Python Application Programming, Prentice Hall,third edition 2015.
	- Deploying Streamlit applications online. Building a GUI-like app with Streamlit for displaying data, charts, and interactive components	**Week 8** (25/08/25-30/08/25)	05	Learning by Demo	Online Forum / Blogs
	Introduction to Flask: Overview of Flask, routing, and creating a simple web application	**Week 9** (01/09/25-06/09/25)	05	**Debugging Challenges**	Wesely J.Chun,Core Python Application Programming, Prentice Hall,third edition 2015.
MID-SEMESTER EXAMINATION Week 10 (08/09/25-13/09/25)					
Unit-4	Writing and Reading Binary	**Week 11** (15/09/25-20/09	05	Lecture-demonstration	Wesely J.Chun,Core Python

FILE HANDLING AND PANDAS	Data, Writing and Parsing Text Files, Writing and Parsing XML Files	/25)		Kahoot, Google classroom flippity **CIA III**	Application Programming ,Prentice Hall,third edition 2015.
	Introduction to Pandas Objects-Data indexing and Selection	**Week 12** (22/09/25-27/09/25)	05	Lecture-demonstration Kahoot, Google classroom	Wesely J.Chun,Core Python Application Programming ,Prentice Hall,third edition 2015.
	Operating on Data in Pandas-Handling Missing Data-Hierarchical Indexing.	**Week 13** (29/09/25-04/10/25)	05	Lecture-demonstration Kahoot, Google classroom flippity	Wesely J.Chun,Core Python Application Programming ,Prentice Hall,third edition 2015.
Unit-5 **NUMPY**	Computation on NumPy-Aggregations-Computation on Arrays-Comparisons	**Week 15** (13/10/25-18/10/25)	05	Lecture-demonstration Kahoot, Google classroom flippity	T.R.Padmanabhan, Programming with Python,Springer Publications,2016
	Masks and Boolean Arrays-Fancy Indexing-Sorting Arrays.	**Week 16** (20/10/25-25/10/25)	05	**Lecture-demonstration Kahoot, Google classroom flippity** /	T.R.Padmanabhan, Programming with Python,Springer Publications,2016
	Structured Data: NumPy's Structured Array. Data Visualization - Different types of Plots.	**Week 17** (27/10/25-01/11/25)	05	Lecture-demonstration Kahoot, Google classroom flippity	T.R.Padmanabhan, Programming with Python,Springer Publications,2016
REVISION		**Week 18** (03/11/25-08/11/25)	**REVISION**		
REVISION		**Week 19** (10/11/25-12/11/25)	**REVISION**		

	ESE Component 1- Regular Program ESE Component 2- Lab Exam			ESE II	

SECTION III

Assessment outline:
This course will follow the assessment outline given below:

CIA - I	CIA – II	CIA – III	Attendance	ESE Component I	ESE Component II
20	50	20	10	50	50

Mapping: A template to map the Learning Outcomes of the course against the components of assessment is given below:

Course Outcomes	Components of assessment				
	CIA - I	CIA – II	CIA – III	ESE I	ESE II
CO 1	YES	YES		YES	YES
CO 2	YES	YES		YES	YES
CO 3		YES	YES	YES	YES
CO4			YES	YES	YES

SECTION IV

Assessment Description: CIA - I

Assessment Name	CIA I (Theory and coding)
Schedule	**Week 5** (04/08/25-09/08/25)
Individual Assignment Details	Individual
Assessment description:	Theory and coding
Mode of the Test	Offline
Marks	20 Marks (10- theory + 10 coding
Duration	60 Minutes

Topics Covered	Unit-1 and Unit 2**Python Basics, Object Oriented Programming Using Python & Regular Expressions:**
Remarks	• Assessment will be based on the execution of scenario-based questions. • Students must complete the evaluation of programs 1 to 5.

Evaluation Rubrics

S. No.	Evaluation Rubrics (20)	Marks
1	Problem Understanding & Logic Design	4
2	Code Implementation & Syntax Accuracy	6
3	Execution & Output Accuracy	4
4	Efficiency & Use of Appropriate Constructs	2
5	Viva / Oral Explanation	4
6	Total	20

Evaluation Rubrics	Marks
R1: Accuracy	✔ Correct Answer 4mark ✔ Wrong Answer 0 mark

Assessment Description: **CIA – III**

Assessment Name	Group based projects
Schedule	**Week 14** (06/10/25-11/10/25) Lab Test
Individual Assignment Details	Individual
Assessment description:	Theory and Lab Test - Scenario based questions 1 / 2.
Mode of the Test	Practical and theory
Marks	20 Marks
Duration	Theory and practical
Topics Covered	Unit 1, 2, and Unit 3 with List of Programs
Remarks	• Assessment will be based on the execution of scenario-based questions.

	● Students must complete the evaluation of programs 1 to 5. ● Evaluation will include both code implementation and oral explanation (viva). ● Academic integrity must be strictly maintained during the practical examination.

Evaluation Rubrics:

Evaluation Rubrics	Maximum Marks	Need to improve	Satisfactory	Proficient
R1: Relevance of the Content	4 Marks	No Relevance [0-1]	Less Relevant [2-3]	Highly Relevant [3-4]
R2: Usage of AI tools	2 Marks	No AI tool used [0-1]	AI Tool used but not properly [1-2]	AI Tool applied efficiently [2]
R3: Time Management	4 Marks	No Time Management [0-1]	Less Time Bound [2-3]	Properly within time limits [3-4]
R4: Clear & Confident Delivery	5 Marks	No Clarity, No Confidence [0-1]	Less Clarity, Low on Confidence [2-3]	Proper Clarity, High on Confidence [4-5]
R5: Viva Voice	5 Marks	No Viva Given[0-1]	Wrong answers[2-3]	Correct answers[4-5]

Assessment Description: ESE – 1

Assessment Name	ESE -1
Schedule	Week-5 (every week in lab)
Individual Assignment Details	Individual
Assessment description:	Regular Lab exercises evaluations
Mode of the Test	Practical & Online Submission
Marks	50 Marks
Topics Covered	UNIT 1-5
Remarks	● Continuous Lab evaluation ● Students need to finish the lab exercise and evaluation on time. ● Each lab document needs to be uploaded in GCR on time. ● Students must follow the lab rules and other instructions given.

Assessment Description: **ESE – 2**

Assessment Name	ESE -2
Schedule	Week-16
Individual Assignment Details	Individual
Assessment description:	Lab test with Viva
Mode of the Test	Lab Test
Marks	50 Marks
Duration	2 Hours
Topics Covered	1 -5 Units (No of questions: 1 / 2 / 3 Scenario-based questions)
Remarks	● Assessment will be based on the execution of scenario-based questions. ● Students must complete the evaluation of all programs from Unit 1-5. ● Evaluation will include both code implementation and oral explanation (viva). ● Academic integrity must be strictly maintained during the practical examination.

Evaluation Rubrics:

S. No.	**Evaluation Rubrics**	**Marks**
1	On-time completion & Domain-based knowledge	10
2	Execution With Complexity	15
3	Formatting & Validation	8
4	Concept Clarity/Completion of task assigned on the spot	10
5	Viva	7
	Total	**50**

Evaluation Rubrics	**Maximum Marks**	**Need to improve**	**Satisfactory**	**Proficient**
R1: Relevance of the Content	10 Marks	No Relevance [0-3]	Less Relevant [3-5]	Highly Relevant [6-10]

R2: Usage of AI tools	10 Marks	No AI tool used [0-3]	AI Tool used but not properly [3-5]	AI Tool applied efficiently [6-10]
R3: Time Management	10 Marks	No Time Management [0-3]	Less Time Bound [3-5]	Properly within time limits [6-10]
R4: Clear & Confident Delivery	10 Marks	No Clarity, No Confidence [0-3]	Less Clarity, Low on Confidence [3-5]	Proper Clarity, High on Confidence [6-10]
R5: Viva Voice	10 Marks	No Viva Given [0-3]	Wrong answers[3-5]	Correct answers [6-10]